

END TERM EXAMINATION [JULY-2023]
SECOND SEMESTER [B.TECH]
APPLIED PHYSICS-II [BS-106]

Time: 3 Hrs.

Max. Marks: 75

Note: Attempt five questions in all including Q. No. 1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q.1. Answer the following questions:-

Q.1. (a) An eigen function of an operator $\frac{d^2}{dx^2}$ is $\psi = e^{ax}$. Find the corresponding Eigen value. (3)

Ans. Refer to Pg. No. 12-2016.

Q.1. (b) Show that Wien's law is a special case of Planck's radiation law. (3)

Ans. Refer to Pg. No. 11-2017.

Q.1. (c) Lattice constant of a cubic lattice is 'a'. Calculate spacing between (011), (101), (112), (111) and (100) planes. (3)

Ans. Spacing between planes are as following.

$$d_{011} = \frac{a}{\sqrt{(0)^2 + (1)^2 + (1)^2}} = \frac{a}{\sqrt{2}}$$

$$d_{101} = \frac{a}{\sqrt{(1)^2 + (0)^2 + (1)^2}} = \frac{a}{\sqrt{2}}$$

$$d_{112} = \frac{a}{\sqrt{(1)^2 + (1)^2 + (2)^2}} = \frac{a}{\sqrt{6}}$$

$$d_{111} = \frac{a}{\sqrt{(1)^2 + (1)^2 + (1)^2}} = \frac{a}{\sqrt{3}}$$

$$d_{100} = \frac{a}{\sqrt{(1)^2 + (0)^2 + (0)^2}} = a$$

Q.1. (d) Explain why semiconductor acts as insulator at OK? (3)

Ans. At OK semiconductor acts as an insulator because at $T = 0K$, $f(E) = 1$, when $E < E_F$, which means all the level below E_F , i.e., valence band are filled up by electron and $f(E) = 0$, when $E > E_F$, i.e., all level above E_F , i.e., conduction bands are empty because at OK, no heat energy is present, so no covalent bands are being broken and the semiconductor behaves as an insulator.

Q.1. (e) The wave function of a particle is $\psi = A \cos^2 x$ for interval $-\pi/2$. Find the value of A. (3)

Ans. Given $\psi = A \cos^2 x$ for interval $-\pi/2$

$$\int_{-\pi/2}^{\pi/2} |\psi|^2 dx = 1$$

$$\int_{-\pi/2}^{\pi/2} A^2 \cos^4 x \, dx = 1$$

$$2A^2 \int_0^{\pi/2} \cos^4 x \, dx = 1$$

$$2A^2 \left(\frac{3\pi}{16} \right) = 1$$

$$A^2 = \frac{16}{3\pi} \times \frac{1}{2} \Rightarrow \boxed{A = \sqrt{\frac{8}{3\pi}}}$$

UNIT - I

Q.2. (a) What do you mean by particle in a box? Show that the energy of an electron in the box varied as the square of the natural numbers. (10)

Ans. Refer to Pg. No. 4-2017.

Q.2. (b) Prove quantum mechanically that particle will not exist in a box if its energy is zero. (3)

Ans. Refer to Pg. No. 23-2018.

Q.2. (c) Differentiate between ψ and $|\psi|^2$. (2)

Ans. The wave function ψ is an important quantity in quantum mechanics measures the wave variation of matter waves. For string waves, the wave variation may be measured by a transverse displacement y , for sound waves, it may be measured by a pressure variation p and for electromagnetic waves, variation may be measured by the variations in electric and magnetic fields. The value of the wave function associated with a moving body at a particular point and time in space is related to the probability of finding the body there.

The wave function ψ cannot be interpreted in terms of an experiment and, therefore has no direct physical significance. Max Born first suggested that the quantity $|\psi|^2$ at any particular point is a measure of the probability that the particle will be near that point. If a volume element dV is constructed around that point, the probability that the particle will be found in the volume element at a given time is $|\psi|^2 dV$. This interpretation of ψ provides a statistical connection between the wave and the associated particle. This tells us where the particle is likely to be not where it is. A large value of $|\psi|^2 dV$ means the strong possibility of the body's presence, while small value means the slight possibility of its presence.

The problem of quantum mechanics is to determine the wave function ψ for a body under action of external forces. Usually wave functions are complex quantity.

However, a probability must be positive and real quantity. The probability density $|\psi|^2$ for complex ψ is taken as product of ψ and its complex conjugate ψ^* . A complex function ψ can be written as:

$$\psi = a + ib \quad (a \text{ and } b \text{ are real})$$

and

$$\psi^* = a - ib$$

Thus

$$|\psi|^2 = \psi \psi^* = a^2 + b^2$$

which is a real quantity.

Q.3. (a) Distinguish between phase and group velocity. Show that the de-Broglie wave group associated with moving particle travels with same velocity as the particle. (10)

Ans.

Phase Velocity	Group Velocity
(i) It is the velocity with which individual wave travels.	It is the velocity with which the resultant envelope of the group of waves travels.
(ii) $V_p = \omega/k$	$V_g = \frac{d\omega}{dk}$

Relation between Group Velocity and Particle Velocity

We know that

Group velocity, $v_g = \frac{d\omega}{dk}$... (i)

Angular frequency $\omega = 2\pi\nu = 2\pi \frac{E}{h}$... (ii)

Differentiating, $d\omega = \frac{2\pi}{h} dE$... (iii)

Propagation constant $k = \frac{2\pi}{\lambda} = 2\pi \cdot \frac{p}{h}$... (iv)

For matter waves, $\lambda = \frac{h}{p}$... (v)

where p = momentum of the particle.

Differentiating E. (v), $dk = \frac{2\pi}{h} dp$... (vi)

Dividing Eq. (iii) by Eq. (vi), we get

$$\frac{d\omega}{dk} = \frac{\frac{2\pi}{h} dE}{\frac{2\pi}{h} dp} = \frac{dE}{dp} \Rightarrow \frac{d\omega}{dk} = \frac{dE}{dp} \quad \dots (vii)$$

We also know that

$$E = \frac{1}{2}mv^2 + V = \frac{1}{2} \frac{m^2 v^2}{m} + V = \frac{1}{2} \frac{p^2}{m} + V \quad [\because V \text{ is constant}]$$

or $dE = \frac{1}{2} \frac{2p}{m} dp = \frac{p}{m} dp$

or $\frac{dE}{dp} = \frac{p}{m}$... (viii)

Since, $p = mv_{\text{particle}}$

$$\frac{dE}{dp} = \frac{mv_{\text{particle}}}{m} = v_{\text{particle}} \quad \dots (ix)$$

From Eqs. (i), (vii) and (ix), we have

$$v_g = v_{\text{particle}} \quad \dots (x)$$

Hence the group velocity of the wave packet is the same as the particle velocity.

Q.3. (b) What is the physical significance of wave function? (3)**Ans.** ψ cannot be interpreted in terms of an experiment. The probability that

something be in certain place at a given time must lie between 0 (the object is not definitely there) and 1 (the object is definitely there). An intermediate probability, say 0.2 means, there is a 20% chance of finding the object. But the amplitude of a wave can be negative as well positive and negative probability is meaningless. Hence ψ by itself cannot be an observed quantity.

The probability of experimentally finding the body described by the wave function ψ at the point (x, y, z) at the time t is proportional to the value of $|\psi|^2$ there at t . A large value of $|\psi|^2$ means the strong possibility of the body's presence, while a small value of $|\psi|^2$, means the slight possibility of its presence. As long as $|\psi|^2$ is not actually zero somewhere, there is a definite chance however small, of detecting it there.

Q.3. (c) The particle trapped in one dimensional box of length L is described by a wave function $\psi = x$. Normalise the wave function between a and b . (2)

Ans. The wave function is-

$$\psi = x$$

Normalized wave function is

$$\int_a^b |\psi|^2 dx = 1$$

$$\int_a^b x^2 dx = 1$$

$$\frac{x^3}{3} \Big|_a^b = 1$$

$$\boxed{b^3 - a^3 = 3}$$

Q.4. (a) State Planck's formula for Black body radiation and derive it from BE statistics. (8)

Ans. Radiator or photon gas is considered as Bose-Einstein gas. Each photon has energy. Let dn be the number of photons distributed in frequency interval ν to $(\nu + d\nu)$.

$$dn = n(\nu)d\nu = g(\nu)f(\nu)d\nu$$

$$g(\nu)d\nu = \frac{8\pi h\nu^2}{c^3} d\nu \quad \dots(1)$$

$$f(\nu) = \frac{1}{(e^{h\nu/k_B T} - 1)} \quad [\text{as } \alpha = 0 \text{ and } E = h\nu] \quad \dots(2)$$

Putting these values in Eq. we get

$$dn = \frac{8\pi h\nu^2}{c^3} \frac{1}{(e^{h\nu/k_B T} - 1)} d\nu \quad \dots(3)$$

Let dE be the energy distributed in frequency interval ν and $(\nu + d\nu)$

$$dE = E dn = h\nu dn \quad \dots(4)$$

From Eqs. (3) and (4), we obtain

$$dE = E(\nu)d\nu = \frac{8\pi h\nu^3}{c^3} \frac{1}{(e^{h\nu/k_B T} - 1)} d\nu \quad (5)$$

Equation (5) is known as Planck's radiation formula for the spectral energy density of black body radiation, which agrees with experimental curve (Fig.)

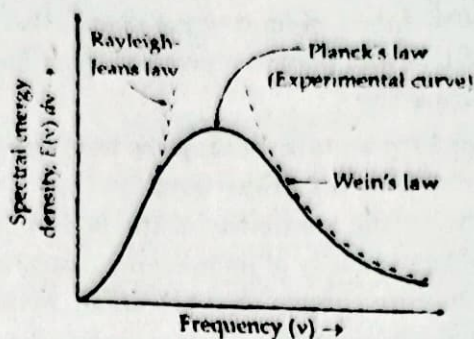


Fig. v vs. $E(v) dv$ curve

UNIT - II

Q.4. (b) If the Sun has a surface temperature of 5700K, what is the wavelength of maximum intensity of solar radiation? (4)

Ans. Given $T = 5700 \text{ K}$

$$\lambda_{\max} T = 2.898 \times 10^{-3} \text{ mk}$$

$$\lambda_{\max} = \frac{2.898 \times 10^{-3}}{5700}$$

$$= 5.084 \times 10^{-7} \text{ m}$$

$$\boxed{\lambda_{\max} = 5084 \text{ \AA}}$$

Q.4. (c) Define Stefan's law. (3)

Ans. Stefan's law is- $E = \sigma T^4$

Energy of radiation is directly proportional to fourth power of absolute temperature of blackbody.

Q.5. (a) An electron gas obeys the Maxwell-Boltzman statistics. Calculate average thermal energy (in eV) of an electron in the system at 300 K. (4)

Ans. Average thermal energy is- $\bar{E} = \frac{3}{2} KT$

Given $T = 300 \text{ K}$

$$\bar{E} = \frac{3}{2} \times \frac{1.38 \times 10^{-23} \times 300}{1.69 \times 10^{-19}} = 388.125 \times 10^{-4}$$

$$\boxed{\bar{E} = 0.03888 \text{ eV}}$$

Q.5. (b) Distinguish between a Boson and Fermions. (3)

Ans. Refer to Pg. 10-2017.

Q.5. (c) What is the relative population of the first two single particle energy levels of a system of distinguishable particles if the energies of the levels is $\epsilon_0 = 0$ and $\epsilon_1 = kT$? (4)

Ans.

$$n_i = \frac{g_i}{e^{E_i/k_B T}}$$

$n_i \rightarrow$ no. of particles

$g_i \rightarrow$ no. of cells

$$\epsilon_0 = E_0 = 0$$

Relative population?

$$\epsilon = E = k_B T$$

$$\alpha = \frac{\mu}{k_B T} \Rightarrow \mu \rightarrow \text{chemical potential}$$

$$n_0 = \frac{g_0}{e^\alpha e^{E_0/k_B T}} = \frac{g_0}{e^\alpha e^0} = \frac{g_0}{e^\alpha} \quad \dots(1)$$

$$n_1 = \frac{g_1}{e^\alpha e^{E_1/k_B T}} = \frac{g_1}{e^\alpha e^{k_B T/k_B T}} = \frac{g_1}{e^\alpha (e^1)} \quad \dots(2)$$

$$\frac{n_1}{n_0} = e^{-1} = 0.367 \quad (g_1 = g_0)$$

Q.5. (d) Discuss the Fermi-Dirac distribution with the help of the distribution function; explain the concept of Fermi level and Fermi energy. (4)

Ans. The F.D. distribution is plotted in Fig. for three different values of T and α . In this distribution the occupation index never goes above 1. This signifies that we cannot have more than 1 particle per quantum state as required by Pauli's exclusive principle which applies in this case. Further, in this distribution the parameter α is strongly dependent on temperature T , and we write

$$\alpha = \frac{-E_F}{k_B T} \quad \dots(1)$$

so that the Fermi-Dirac occupation index becomes

$$f_{FD}(E_i) = \frac{1}{(e^{(E_i - E_F)/k_B T} + 1)} \quad \dots(2)$$

where E_F is called Fermi energy

Let us consider the situation at the absolute zero of temperature. At $T = 0$, $(E_i - E_F)/k_B T = +\infty$ (for $E_i < E_F$) and $(E_i - E_F)/k_B T = -\infty$ (for $E_i > E_F$).

Therefore

$$\text{for } E_i < E_F \quad f_{FD}(E_i) = \frac{1}{e^{-\infty} + 1} = 1 \quad (\because e^{-\infty} = 0) \quad \dots(3)$$

$$\text{and for } E_i > E_F \quad f_{FD}(E_i) = \frac{1}{e^{+\infty} + 1} = 0 \quad (\because e^{+\infty} = \infty) \quad \dots(4)$$

Thus at $T = 0$, all energy states from $E_i = 0$ to $E_i = E_F$ are occupied because $f_{FD}(E_i) = 1$, while all states above E_F are vacant.

As the temperature rises, some of the states just below E_F become vacant, while some just above E_F are occupied. The higher the temperature, the more in the spreading in $f_{FD}(E_i)$.

At $E_i = E_F$, we have

$$f_{FD}(E_i) = \frac{1}{e^0 + 1} = \frac{1}{2}, \text{ at all temperatures.} \quad \dots(5)$$

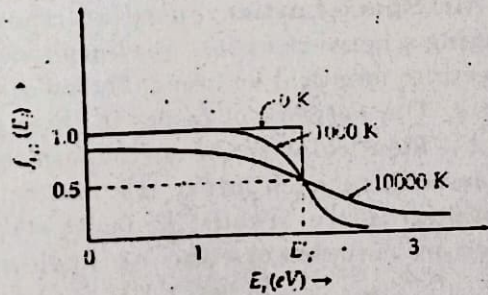


Fig. F.D. distribution curve.

That is, the average number of particles per quantum state is exactly $\frac{1}{2}$. In other words, the probability of finding an electron with energy equal to the Fermi energy in a metal is $\frac{1}{2}$ at any temperature.

Q.6. (a) Why do we use X-ray to study crystal structure? (3)

Ans. Refer to Pg. No. 11-2019.

Q.6. (b) State Bragg's law? How is it applied? (3)

Ans. Refer to Pg. No. 11-2019.

Q.6. (c) Write short notes on: (9)

(i) Point defects

(ii) Frenkel defects

(iii) Schottky defects

Ans. Refer to Pg. No. 15-2016.

Q.7. (a) Define the following:- (8)

(i) Unit Cell

(ii) Space Lattice

(iii) Coordiante number

(iv) Miller indices?

Ans. (i) Unit cell: Refer to Pg. No. 13-2016.

(ii) Space Lattice: Space lattice is a geometrical abstraction. To have its meaning, imaging a network of straight lines constructed in such a way that it divides the entire space into identical volumes. The points of intersection of these lines are called lattice points. The network of points in three dimensions, in which the surrounding of each point is identical with the surrounding of other points, is called space lattice (Fig. 3(a)). By associating each lattice point with a single atom or group of atoms, called basis, crystal structure- results. Thus, for every crystal, there is a network of lattice points, which are occupied by either s a single atoms or group of atoms.

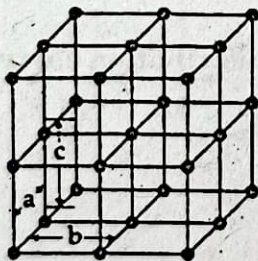


Fig. 3(a) Space lattice and fundamental spanned by vectors a, b and c.

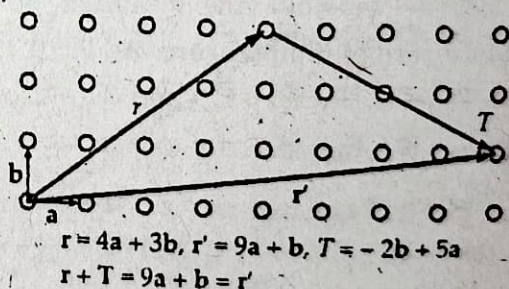


Fig. 3 (b) 2D space lattice basis vectors a and b.

In 1848, a Russian mathematician Bravais showed that there are just 14 ways of arranging points in space lattice such that all the lattice points have exactly the same surroundings. Such lattices are called Bravais lattices². From these 14 space lattices, unlimited number of different crystal structures can be made. Two dimensional space lattice spanned by basis vectors a and b is shown in Fig. 3 (b).

For any type of lattice, there exist three fundamental translational vectors a, b and c, not lying in a plane, in terms of which any lattice point can be specified. A translation operation T is defined as

$$T = n_1 a + n_2 b + n_3 c \quad \dots(1)$$

where n_1, n_2, n_3 be integers. Thus it is meant that by applying the operation T to a point r , we reach another point r' , which has the same environment as r .

$$r' = r + T = r + n_1 a + n_2 b + n_3 c \quad \dots(2)$$

The set of points r , given by Eq. (2) for all possible values of the integers n_1, n_2 , and n_3 defines the space lattice.

(iii) **Coordination Number:** It is the number of nearest neighbouring atoms molecules or ions to a particular atom.

(iv) **Miller indices:** The crystal lattice may be regarded as made up of an infinite set of parallel equidistant planes passing through the lattice points which are known as lattice planes. In simple language, we can say that the plane passing through lattice points are called "lattice planes". For a given lattice, the lattice planes can be chosen in a different number of ways.

For example, four planes are shown in Fig. (4).

The orientation of planes or faces in a crystal can be described in terms of their intercepts on the three axes. Miller introduced a system to designate a plane in a crystal. He introduced a set of three numbers to specify a plane in a crystal. This set of three numbers is known as 'Miller indices' of the concerned plane.

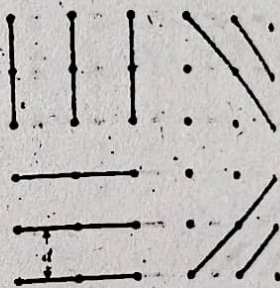


Fig. (4) Different lattice planes

Definition

Miller indices are defined as the reciprocals of the intercepts made by the plane on the three axes.

Procedure for Finding Miller Indices

Steps :

1. Determine the intercepts of the plane along x, y and z axes in terms of the lattice constants a, b and c .
2. Determine the reciprocals of these numbers
3. Find the least common denominator (LCD) and multiply each by this LCD.
4. The result is written in parenthesis.

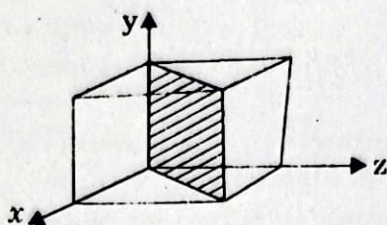
This is called 'Miller indices' of the plane in the form (hkl) . $\uparrow$ z

Q.7. (b) Deduce Miller indices for the plane having intercepts a, b and c at $-2, 1, -2$. Also draw the plane. (4)

Ans. Intercepts at $-2, 1, -2$

Reciprocal are- $-\frac{1}{2}, \frac{1}{1}, -\frac{1}{2}$

LCD - -1, 0, -1

Miller indices for the plane ($\bar{1}01$)

Q.7. (c) X-rays of wavelength 2×10^{-11} m suffer first order reflection from (111) crystal plane at an angle of 45° . What is the inter atomic spacing of the crystal? (3)

Ans. Given

$$\lambda = 2 \times 10^{-11} \text{ m}$$

$$n = 1$$

$$\theta = 45^\circ$$

plane - (111) = (hkl)

Inter-atomic spacing is - "a"

We know,

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

$$a = d\sqrt{h^2 + k^2 + l^2}$$

But $2d \sin \theta = n\lambda$

$$d = \frac{n\lambda}{2\sin \theta}$$

$$a = \frac{n\lambda}{2\sin \theta} \times \sqrt{h^2 + k^2 + l^2}$$

$$a = \frac{1 \times 2 \times 10^{-11}}{2 \times \sin 45^\circ} \times \sqrt{1^2 + 1^2 + 1^2}$$

$$a = \frac{2 \times 10^{-11}}{2 \times \frac{1}{\sqrt{2}}} \times \sqrt{3} = \sqrt{6} \times 10^{-11}$$

$$a = 2.44 \times 10^{-11} \text{ m}$$

UNIT - IV

Q.8. (a) Distinguish between intrinsic and extrinsic semiconductors. (4)

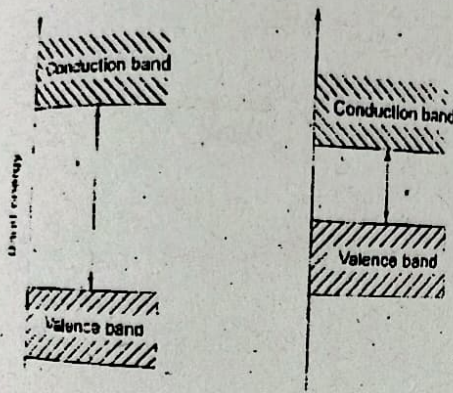
Ans. Refer to Pg. No. 16-2017.

Q.8. (b) Explain Kronig-Penney model for the motion of electron in a periodic potential. (7)

Ans. Refer to Pg. No. 20-2016.

Q.8. (c) Differentiate conductor, insulator and semiconductor using energy-band diagram. (4)

Ans.



(a) Insulator (b) Semiconductor (c) Metal

Fig. Energy band model of insulator, semiconductor and metal.

Q.9. (a) Show that the Fermi energy lies midway between the conduction band and valence band for intrinsic semiconductor. (6)

Ans. Refer to Pg. No. 25-2016 and

For diagram: Refer to Pg. No. 24-2016. (Fig. 11(a).

Q.9. (b) Write short notes on:-

(3 + 3 + 3 = 9)

(i) Zener diode

(ii) PN junction diode

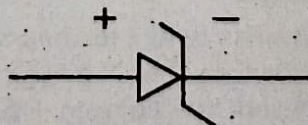
(iii) Photodiode

Ans. (i) Zener diode

It is a reverse biased heavily doped p-n junction diode which is operated in the breakdown region. or A properly doped crystal diode having a sharp breakdown voltage is known as a zener diode.

The ordinary diodes destroy and cannot work in the breakdown region. But zener diodes are properly doped. When doping is large, depletion layer is very narrow. The reverse voltage produces an intense electric field across the depletion layer producing extremely large no of electron-hole pairs. This results in a sharp increase in reverse current called the reverse saturation current whose value is limited by the external resistance in the circuit. It is independent of the applied voltage. The sudden increase in current is known as zener current. The voltage across the zener diode is almost constant for a large change of current through it. This voltage is called zener voltage.

The symbol of zener diode is:



$$I = I_s [e^{eV/k_B T} - 1]$$

...(1)

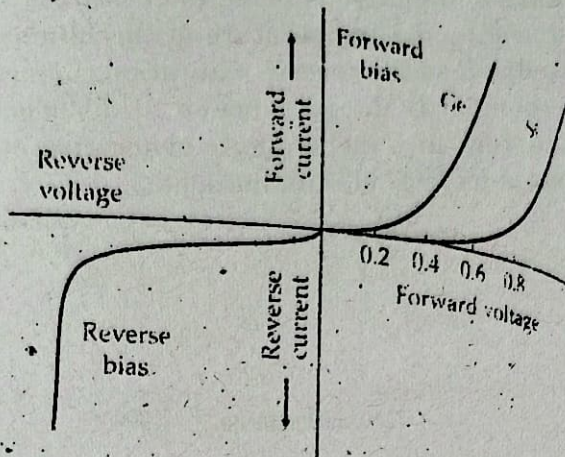


Fig. 10 Characteristics curve of a P-N junction

where I_s is equal to the saturation current under reverse bias, e is the electric charge, k_B the Boltzmann's constant, T the temperature in K and V is the bias voltage applied to the diode. The term $k_B T/e$ in Eq. (1) has the dimensions of voltage and is known as the equivalent of temperature, denoted by V_T .

(iii) **Photodiode:** It is a two terminal semiconductor pn junction and is designed to operate with reverse bias. The photodiode is a photovoltaic device as well as photoconductive device. When the device operates with reverse bias, it operates as a photoconductive device and when operates without reverse bias, it operates as a photodiode device.

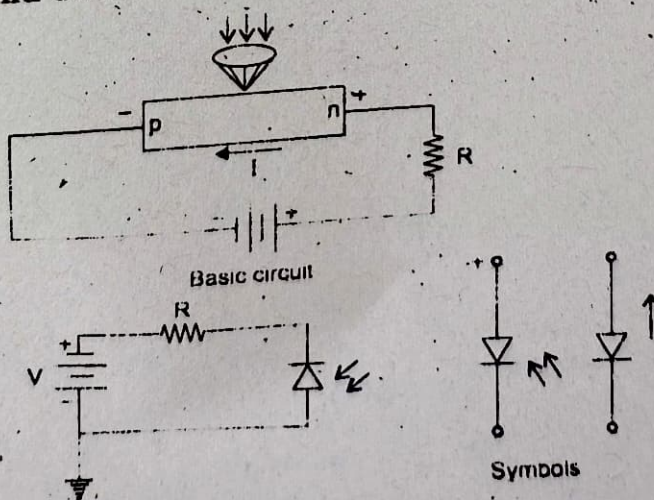
The operating wavelength depends on the band-gap energy of the material. The material mostly used in photodiodes are semiconductor like Si and Ge. Group III-V and II-IV compound semiconductors like GaAs, InP and CdTe are also used. Photodiode is composed of...

Si - Used in consumer electronics

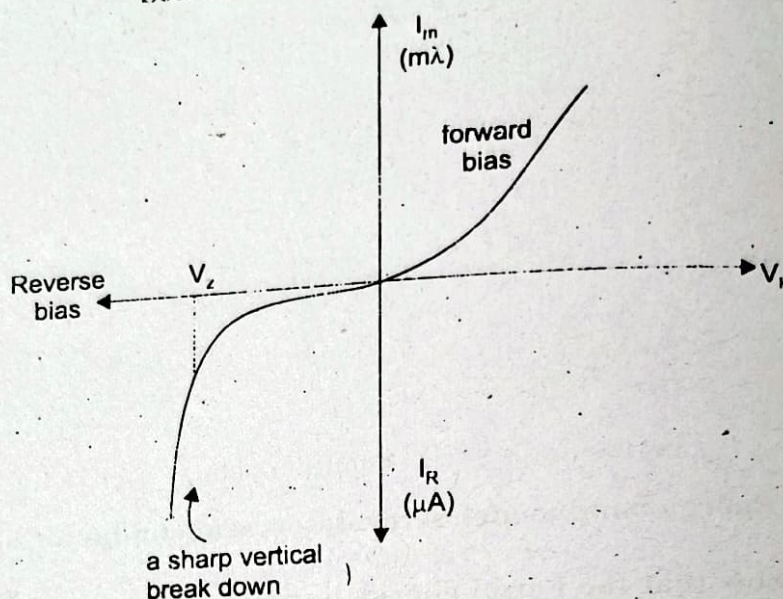
Ge and InGaAs (P)/InP - Used in optical fibre

InAs and InSb (longer λ)-

Construction and Characteristics of Photodiodes-



18-2023



Properties of a zener diode:

1. It is properly doped to have a sharp & vertical breakdown.
2. It operates in reverse biased condition.
3. In forward bias, its characteristics are similar to that of an ordinary diode.
4. Zener diode is designed such that it is not destroyed at breakdown region.
5. It maintains a constant output voltage even through the current changes through at breakdown region. Hence, it is used as a voltage regulator.
6. Zener breakdown occurs at less reverse voltages while Avalanche breakdown occurs at higher reverse voltages.

(ii) P-N junction diode-

Most semiconductor diodes consist of a P-N junction. The circuit symbol for a normal diode is shown in Fig. 9 the arrow indicates the forward direction of the diode. The diffusion of the majority charge carriers across the P-N junction sets up a potential barrier at the junction with P-side as negative and N-side as positive. This barrier is just sufficient to stop the movement of carriers across the junction. If now an external potential difference with a polarity so as to oppose the barrier is applied as shown in Fig. 9 (b), movement of majority carriers across the junction would be resumed a current will flow through the junction in the P to N direction.

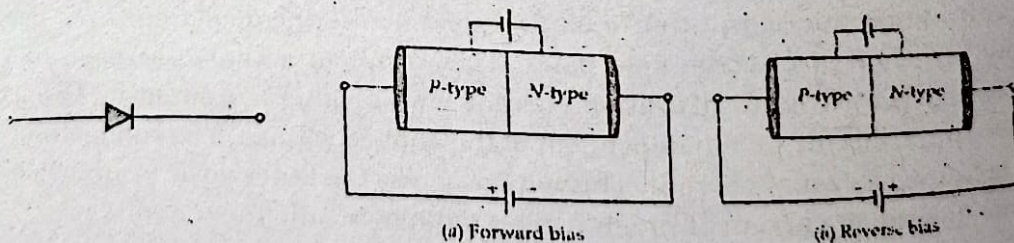


Fig.9(a) Circuit symbol of a normal semiconductor diode

Fig. 9(b) Biasing of a P-N junction

The external voltage is then known as the forward bias voltage, the resulting current as the forward current and the direction P to N as the forward direction of the diode. As the forward bias voltage is increased, the current I through the junction increases as shown in Fig. 10. The increase in current is nearly exponential for the most part and it follows an equation known as the diode equation,

MID TERM EXAMINATION [MAY-2023]
SECOND SEMESTER [B.TECH]
APPLIED PHYSICS-II [BS-106]

Time: 1:5 Hrs.

Max. Marks: 30

Note: Attempt Question no.1 which is compulsory and any two more questions from remaining.

Q.1. (a) Distribute two particles in three energy states in accordance of MB, BE and FD statistics. (2.5)

Ans. For Maxwell-Boltzmann (MB) statistics, the two particles are considered as distinguishable. We shall denote them as A and B and the quantum states as 1, 2, 3. In MB statistics any number of particles can occupy the same quantum state. The possible states of the two-particle system are shown in the table below:

1	2	3
AB		
	AB	
		AB
A	B	
B	A	
	A	B
	B	A

For Bose-Einstein (BE) statistics the two particles are considered to be indistinguishable. We shall denote them both as A. In BE statistics, any number of particles can occupy the same quantum state. The possible states of the two-particle system are shown in the table below :

1	2	3
AA		
	AA	
		AA
A	A	
	A	A
A		A

We can see that there are six distinct states available to the two-particle system.

For Fermi-Dirac (FD) statistics, the two particles are considered to be indistinguishable. We shall denote them both as A. In FD statistics, no more than one particle can occupy a given quantum state.

1	2	3
A	A	
	A	A
A		A

The number of available distinct states are now reduced to three.

Q.1. (b) What do you understand by normalisation of a wave function? Give the essential features of acceptable wave function. (2.5)

Ans. Refer to (Pg. No. 22-2018 and Pg. 15-2017).

Q.1. (c) Determine the uncertainty in position of an electron whose speed is measured to be 5.0×10^3 m/sec to an accuracy of 0.003%. (2.5)

Ans.

$$\Delta x \cdot \Delta p_x = \frac{h}{2\pi}$$

$$\Delta x = \frac{h}{2\pi \Delta p_x} = \frac{h}{2\pi m \Delta V_x}$$

Given,

$$\Delta V = 5000 \text{ m/s} \times \frac{0.003}{100} = 0.15 \frac{\text{m}}{\text{s}}$$

$$\Delta x = \frac{6.63 \times 10^{-34}}{2 \times 3.14 \times 9.1 \times 10^{-31} \times 0.15}$$

$$\Delta x = 7.74 \times 10^{-4} \text{ m} \quad \text{Ans.}$$

Q.1. (d) Compute the root mean square speed of oxygen molecules at 0°C . Given mass of $\text{O}_2 = 5.31 \times 10^{-26}$ kg. (2.5)

Ans.

$$V_{\text{rms}} = \sqrt{\frac{3K_B T}{m}}$$

$$= \sqrt{\frac{3 \times 1.38 \times 10^{-23} \times 273}{5.31 \times 10^{-26}}} = 461.35 \frac{\text{m}}{\text{s}}$$

Q.2. (a) Evaluate the expectation value of the position and the momentum of a particle trapped in a box of width L . (5)

Ans.

$$\langle p_x \rangle = \int_{-a}^a \psi^* \left(-i\hbar \frac{\partial}{\partial x} \right) \psi dx$$

$$= \frac{2}{L} (-i\hbar) \int_0^L \sin\left(\frac{n\pi x}{L}\right) \cdot \frac{\partial}{\partial x} \sin\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{2}{l}(-i\hbar) \int_0^l \sin\left(\frac{n\pi x}{l}\right) \cdot \frac{n\pi}{l} \cos\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{2n\pi}{l^2}(-i\hbar) \int_0^l \sin \frac{n\pi x}{l} \cos \frac{n\pi x}{l} dx = 0$$

Q.2. (b) Determine the energy required to jump from ground state to second excited state by an electron trapped in a box of length 1 Å. (2)

Ans. $E_n = \frac{n^2 h^2}{8ml^2}$

$$\Delta E = E_3 - E_1 = \frac{(9-1) \times (6.623 \times 10^{-34})^2}{8 \times (9.1 \times 10^{-31}) \times (10^{-10})^2}$$

$$\Delta E = 4.8 \times 10^{-17} \text{ J}$$

$$\Delta E = \frac{4.8 \times 10^{-17}}{1.6 \times 10^{-19}}$$

$$\boxed{\Delta E = 300 \text{ eV}} \text{ Ans.}$$

Q.2. (c) Brief the conditions required for the quantum tunnelling. Calculate the transmission probability for an electron of energy 2eV when incident upon a rectangular potential barrier of height 6eV and width 0.50 nm. (3)

Ans. $T = \frac{16E}{V_0} \left[1 - \frac{E}{V_0} \right] e^{-2\beta a}$

Given, $E = 2\text{eV}$, $V_0 = 6\text{eV}$, $\hbar = \frac{h}{2\pi} = 1.054 \times 10^{-34} \text{ J}$

$a = 0.5 \text{ nm} = 0.5 \times 10^{-9} \text{ m}$, $m = 9.1 \times 10^{-31} \text{ kg}$

$$2\beta a = \frac{2a[2m(V_0 - E)]^{1/2}}{\hbar / 2\pi}$$

$$= \frac{2 \times 0.5 \times 10^{-9} [2 \times 9.1 \times 10^{-31} (6-2) \times 1.6 \times 10^{-19}]^{1/2}}{1.054 \times 10^{-34}}$$

$$= 8.87$$

$$T = \frac{16 \times 2}{6} \left[1 - \frac{2}{6} \right] e^{-8.87}$$

$$\boxed{T = 0.000497}$$

Barrier Potential (Tunneling Effect)

Let us consider the one-dimensional problem, where the potential function is defined as

$$\left. \begin{aligned} V(x) &= 0 \text{ for } x < 0 \\ &= V_0 \text{ for } 0 < x < a \\ &= 0 \text{ for } x > a \end{aligned} \right\} \dots(1)$$

Here, we have a potential between $x = 0$ and $x = a$ (Fig. 1).

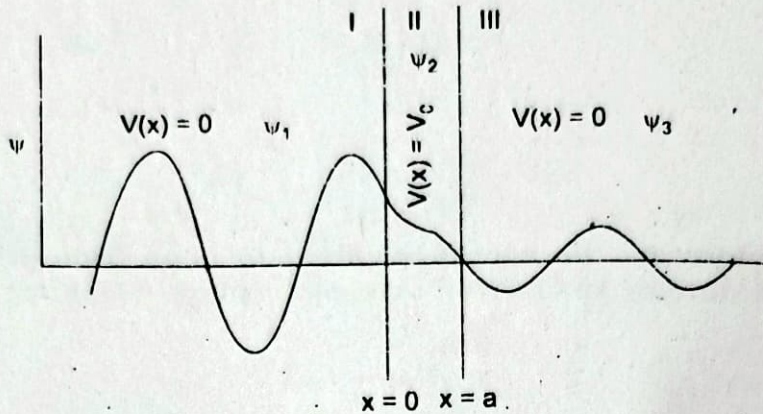


Fig.1 Barrier Potential

If a particle having energy less than V_0 i.e., $E < V_0$ approaches this barrier from the left i.e., from first region, classically the particle will always be reflected and hence will not penetrate the barrier. However the quantum mechanics predicts that the particle has some probability of penetrating to region III, the probability of penetration being greater if $(V_0 - E)$ and a are smaller. Moreover, if $E > V_0$, classical mechanics predicts that the particle will always be transmitted; while according to quantum-mechanics, the particle has a finite probability of transmission as well as reflection.

Q.3. (a) Find the expression of the average molecular energy of an ideal gas molecule. (4)

Ans. The Maxwell-Boltzmann distribution law is

$$n_i = \frac{g_i}{e^{(u \cdot E_i / k_B T)}} \quad \dots(1)$$

Consider an ideal gas that contains N molecules. Consider a continuous distribution of molecular energies instead of the discrete set $E_1, E_2, E_3, \dots$ then Eq. (1) becomes.

$$n(E)dE = g(E)e^{-E/k_B T} dE \quad \dots(2)$$

Here $n(E)dE$ represents the number of molecules having energies between E and $(E+dE)$. $g(E)dE$ represents the number of states that have energies between E and $E + dE$.

Let us first find $g(E)dE$. A molecule of energy E has a momentum p whose magnitude is

$$p = \sqrt{2mE} = \sqrt{p_x^2 + p_y^2 + p_z^2} \quad \dots(3)$$

Each set of momentum components p_x, p_y, p_z specifies a different state of motion. Let us imagine a momentum space whose coordinate axes are p_x, p_y and p_z (Fig. 2). The number of momentum states available to a particle with a momentum whose magnitude is between p and $(p + dp)$ is proportional to the volume of a spherical shell in momentum space of radius p and thickness dp . Volume of this spherical shell is $4\pi p^2 dp$.

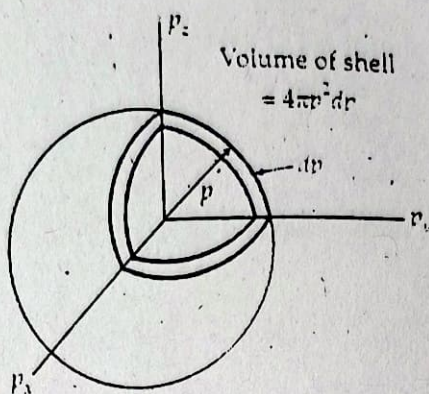


Fig. 2 Momentum Space.

Hence

$$g(p)dp = Bp^2dp \quad \dots(4)$$

where B is some constant. Since each momentum magnitude p corresponds to a single energy E , the number of energy states $g(E)dE$ between E and $(E + dE)$ is the same as the number of momentum states $g(p)dp$ between p and $(p + dp)$.

$$g(E)dE = Bp^2dp \quad \dots(5)$$

But $p^2 = 2mE$ and $dp = \frac{2mdE}{\sqrt{2mE}}$

$$g(E)dE = \sqrt{2m}^{3/2} B \sqrt{E} dE \quad \dots(6)$$

Substituting for $g(E)dE$ in Eq. (2)

$$n(E)dE = C \sqrt{E} e^{-(E/k_B T)} dE \quad \dots(7)$$

where $C = \sqrt{2m}^{3/2} B e^{-1} = \text{constant}$.

To find C we make use of the normalisation condition that the total number of molecules is N .

Therefore
$$N = \int n(E)dE = C \int_0^\infty \sqrt{E} e^{-(E/k_B T)} dE \quad \dots(8)$$

From definite integrals as given below

$$\int_0^\infty \sqrt{x} e^{-ax} dx = \frac{1}{2a} \sqrt{\frac{\pi}{a}}$$

We find that

$$N = C \frac{1}{2(1/k_B T)} \sqrt{\frac{\pi}{(1/k_B T)}} = C \times \frac{\sqrt{\pi}}{2} \times (k_B T)^{3/2} \quad \dots(9)$$

or

$$C = \frac{2\pi N}{(\pi k_B T)^{3/2}}$$

$$n(E)dE = \frac{2\pi N}{(\pi k_B T)^{3/2}} \sqrt{E} e^{-E/k_B T} dE \quad \dots(10)$$

Equation (10) is plotted in Fig. 2 in terms of $k_B T$

The total internal energy of the system is

$$E = \int_0^{\infty} E n(E) dE = \frac{2\pi N}{(\pi k_B T)^{3/2}} \int_0^{\infty} E^{3/2} e^{-E/k_B T} dE$$

The value of the definite integral is $\frac{3}{4} (k_B T)^2 \sqrt{\pi k_B T}$

$$\therefore E = \frac{2\pi N}{(\pi k_B T)^{3/2}} \times \frac{3}{4} (k_B T)^2 \sqrt{\pi k_B T} = \frac{3}{2} N k_B T \quad \dots(11)$$

The average energy of an ideal gas molecule is (E/N)

$$\therefore E = \frac{3}{2} k_B T \quad \dots(12)$$

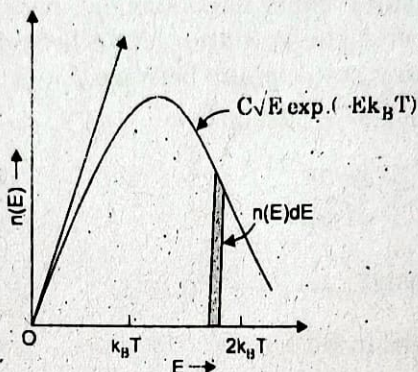


Fig. 3

Q.3. (b) How does Fermions differ from Bosons? Give examples. (3)

Ans. Refer to (Pg. No. 10-2017)

Q.3. (c) Calculate the average energy and speed of a free electron at 0K for a given substance? Given fermi energy of substance is 7.9 eV. (3)

Ans. Average Energy is

$$\bar{E} = \frac{3}{5} E_F$$

$$\bar{E} = \frac{3}{5} \times 7.9 = 4.74 \text{ eV} \Rightarrow \boxed{\bar{E} = 4.74 \text{ eV}}$$

Average speed

$$\bar{E} = \frac{1}{2} m v^2$$

$$v = \sqrt{\frac{2\bar{E}}{m}} = \sqrt{\frac{2 \times 4.74 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}}}$$

$$\boxed{v = 1.2 \times 10^6 \frac{\text{m}}{\text{sec}}}$$

Q.4. (a) Using Schrodinger's equation, find wave function for a particle of mass m in an infinite potential well of width L . Also find its eigen energy values. (6)

Ans. Refer to (Pg. No. 5-2019).

Q.4. (b) An electron gas obeys the Maxwell-Boltzmann statistics. Calculate the average thermal energy (in eV) for an electron in the system at 300K. (2)

Ans.

$$E = \frac{3}{2} KT$$

$$E = \frac{3}{2} \times 1.38 \times 10^{-23} \times 300 \text{ J}$$

$$E = \frac{3 \times 1.38 \times 10^{-23} \times 300}{2 \times 1.6 \times 10^{-19}}$$

$$E = 0.039 \text{ eV}$$

Q.4. (c) States the Plancks' radiation law and give its expression for energy density. (2)

Ans. Refer to (Pg. No. 2-2019).

Study hard

